In-Context Gibbs: Learning from Weak Labels with Frozen Tabular Foundation Models

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Abstract

[TODO: abstract, written last]

1 Introduction

Contributions.

- **Method** (Section 4): In-Context Gibbs, a training-free sampler over latent labels that uses a frozen TFM as its full conditional and handles instance-level (PLL) and group-level (LLP) weak labels with one procedure [C2].
- **Theory** (Section 5): exactness when the TFM is the posterior predictive of its prior [C3]; stability when it is only approximate, even if its conditionals are incompatible [C4]; and a characterisation of when the naive use of a TFM is already sufficient [C12].
- **Evidence from two pre-registered studies** (Section 6): with TabPFN v2 (Study 1) the method did not beat the strongest specialised baselines; a pre-registered replication with TabICL v2 on 28 new datasets (Study 2) did, on both LLP and PLL (Table 1) [C6][C8].
- **Mechanism** (Section 6.5): the advantage comes from modelling dependence between latent labels through the context [C7], and whether it materialises depends on how robust the TFM is to noisy context labels [C13].

[TODO: intro prose]

2 Related work

[TODO: related work prose and Table 1 (positioning)]

3 Problem setup

[TODO: formal setup]

Table 1: Pre-registered confirmatory tests. Δ: mean over datasets of (In-Context Gibbs $-$ comparator), each dataset averaged over 3 seeds; W/T/L: datasets won/tied/lost; p_{Holm}: two-sided Wilcoxon signed-rank, Holm-adjusted within each hypothesis (2 tests for LLP, 4 for PLL). Bold: significant in favour of In-Context Gibbs. "Naive TFM" uses the same TFM as the study. Study 1: 10 binary and 14 multiclass datasets; Study 2: 18 binary and 10 multiclass datasets disjoint from Study 1. H2/H4 additionally required winning against the strongest baseline on $\geq$70%/$\geq$60% of datasets (Section 6.4).

			Study 1: TabPFN v2			Study 2: TabICL v2, new datasets		
Setting	Metric	Gibbs vs.	Δ	W/T/L	p_{Holm}	Δ [95% CI]	W/T/L	p_{Holm}
LLP, bag size 8	AUC	DLLP (linear)	+0.024	8/0/2	0.232	+0.052 [+0.024, +0.086]	16/0/2	$<$**0.001**
		Naive TFM	+0.148	10/0/0	**0.004**	+0.116 [+0.085, +0.150]	18/0/0	$<$**0.001**
PLL, $q = 0.3$	Acc.	CC (tuned)	+0.022	10/0/4	0.084	+0.012 [+0.006, +0.019]	8/2/0	**0.016**
		Naive TFM	+0.099	13/0/1	$<$**0.001**	+0.022 [+0.012, +0.033]	10/0/0	**0.008**
PLL, $q = 0.6$	Acc.	CC (tuned)	$-$0.132	7/0/7	0.715	+0.012 [+0.006, +0.019]	8/1/1	**0.016**
		Naive TFM	+0.180	12/0/2	**0.002**	+0.205 [+0.094, +0.337]	10/0/0	**0.008**
Pre-registered verdict			H2: not met; H4: not met			H2′: met; H4′: met		

4 In-Context Gibbs

[TODO: method prose and Algorithm 1]

5 Theory

[TODO: theory section]

6 Experiments

6.1 Design: two pre-registered studies

[TODO: design prose; pre-registration timeline table in Appendix]

6.2 Correctness on problems with an exact posterior

[TODO: RQ1 paragraph]

6.3 Study 1: TabPFN v2

Study 1 evaluated the frozen protocol with TabPFN v2 [28] as the conditional (Table 1, left). In-Context Gibbs improved on the naive use of the same TFM in both main settings, for example by +0.148 AUC on LLP, but it did not beat the strongest specialised baselines. On LLP with bags of 8 it was +0.024 AUC ahead of linear DLLP (8/0/2, $p_{\text{Holm}} = 0.232$). On PLL it was ahead of tuned CC at $q = 0.3$ (+0.022, $p_{\text{Holm}} = 0.084$) and far behind at $q = 0.6$ ($-$0.132), where the chains collapsed on the four datasets with at least eight classes (Figure 2, bottom). Both pre-registered main hypotheses therefore failed; the mechanism hypothesis H3 held (Table 2).

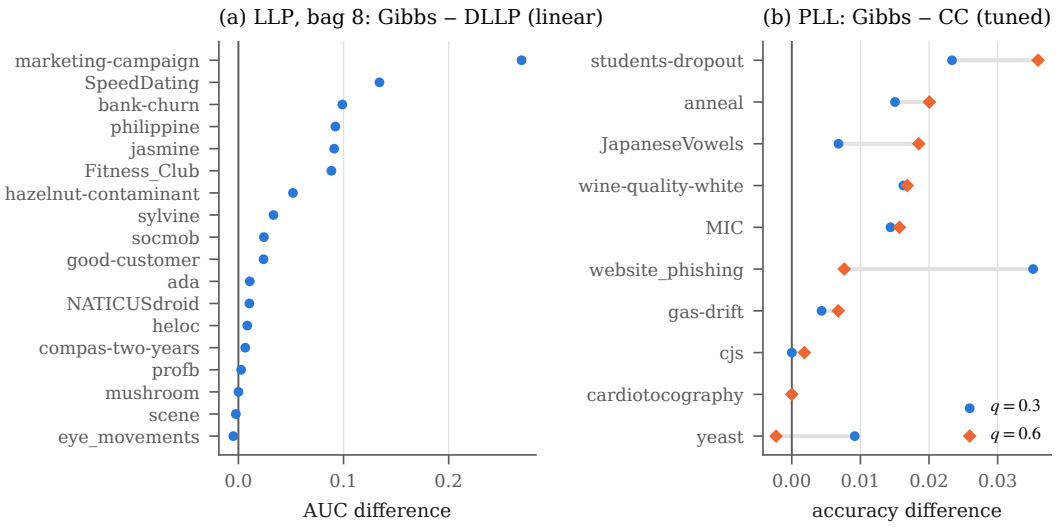


Figure 1: Study 2, per dataset: difference between In-Context Gibbs and the primary comparator (positive favours In-Context Gibbs), mean over 3 seeds. (a) LLP with bags of 8, AUC against linear DLLP. (b) PLL, accuracy against tuned CC at two candidate-set rates; a horizontal bar joins the two rates of a dataset.

6.4 Study 2: a pre-registered replication with TabICL v2 on new datasets

In an ablation of Study 1, replacing TabPFN v2 by TabICL v2 [7] passed every test (Appendix C). Because that TFM was chosen after seeing the results, we treated the finding as a hypothesis and tested it in Study 2, changing nothing but the TFM, on 28 datasets disjoint from those of Study 1. All pre-registered tests passed (Table 1, right; Figure 1). On LLP with bags of 8, In-Context Gibbs improved AUC over linear DLLP by +0.052 [+0.024, +0.086] (16/0/2, $p_{\text{Holm}} < 0.001$) and beat the strongest LLP baseline, tuned DLLP (MLP), on 16/18 datasets (required: $\geq$70%) [C6]. On PLL it improved accuracy over tuned CC by +0.012 at $q = 0.3$ (8/2/0) and by +0.012 at $q = 0.6$ (8/1/1; $p_{\text{Holm}} = 0.016$ for both), and was at least as accurate as the strongest PLL baseline on 10/10 and 9/10 datasets (required: $\geq$60% at each level) [C8]. Unlike in Study 1, it did not collapse on the four datasets with at least eight classes. The PLL gains are modest in absolute terms: In-Context Gibbs is within 0.8 and 1.9 accuracy points of an oracle trained on the true labels, so the room for improvement is small. Removing datasets flagged for group structure or temporal drift did not change any conclusion (Appendix D).

6.5 Why it works, and when it does not

Sampling versus hard assignment. With the same TFM and the same warm start, In-Context Gibbs had lower log loss than hard-EM in 86.7% of 128 cells in Study 1 and 78.9% of 38 cells in Study 2, and the advantage grew with the weakness of the labels (Table 2) [C7].

Robustness of the TFM to noisy context labels. Figure 2 (top) measures what one Gibbs sweep does to an assignment in which a fraction r of the training labels is correct (PLL, $q = 0.6$; exploratory, Study 1 datasets only). With TabPFN v2 the gain $\Phi(r) - r$ is negative over a range of r on all four datasets with at least eight classes, so a chain that starts near the uninformative state is pushed back towards it; with TabICL v2 the gain is positive everywhere. The chains behave accordingly (bottom): TabPFN chains stay near chance on mfeat-factors and elsewhere escape slowly, or not at all, depending on the seed, whereas TabICL chains converge within about ten sweeps [C13]. [TODO: link to the collapse-threshold analysis (G9_PLAN D2), if a proposition is ready by 20/12]

Table 2: In-Context Gibbs versus hard-EM with the same TFM and warm start, log loss. A cell is a (dataset, weak-label level) pair; Bayesian signed-rank test with ROPE ± 0.005; Spearman correlation between weakness level and the gap. †Computed on PLL cells (LLP has a single level), as stated before the analysis was run.

Study (TFM; settings)	cells	Gibbs better	P(Gibbs better)	P(ROPE)	Spearman ρ
Study 1 (TabPFN v2; all six settings)	128	86.7%	1.00	0.00	0.36
Study 2 (TabICL v2; LLP bag 8, PLL)	38	78.9%	1.00	0.00	0.66†

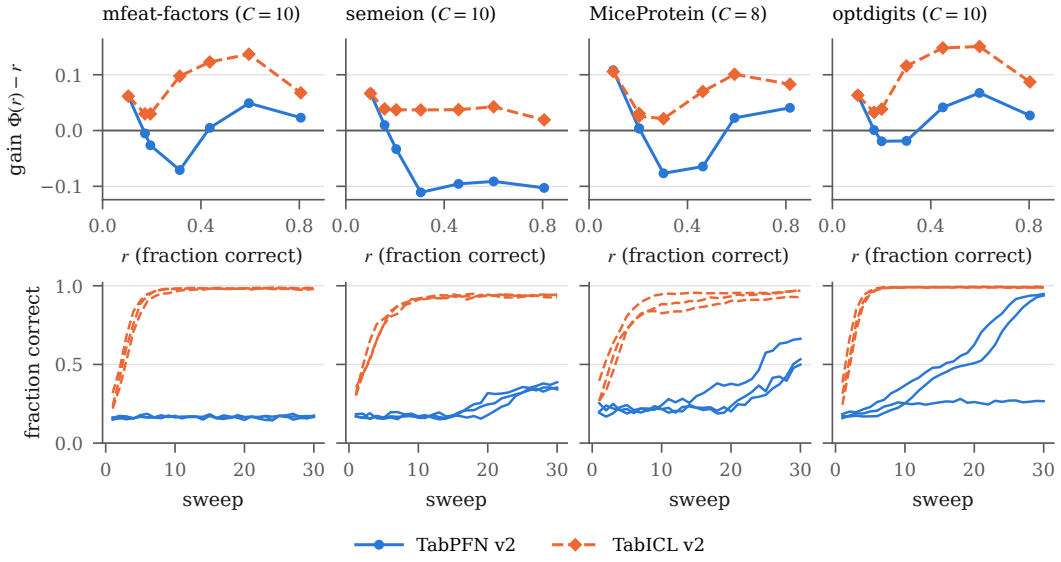


Figure 2: PLL at $q = 0.6$ on the four Study 1 datasets with at least eight classes. Top: gain $\Phi(r) - r$ of one Gibbs sweep started from an assignment with a fraction r of correct labels (mean over 3 seeds; exploratory). Bottom: fraction of correct training labels along the chains of the main runs, one line per seed (diagnostic only; the sampler never sees the true labels).

6.6 Real weak labels

[TODO: real-PLL paragraph and table (numbers via make_tables.py)]

6.7 Ablations and cost

[TODO: ablation and cost paragraph]

7 Limitations

[TODO: limitations prose]

8 Conclusion

[TODO: conclusion: only claims already supported in the paper]

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A Proofs

[TODO: from G4/PROOFS.md]

B Pre-registration and timeline

[TODO: timeline table]

C Study 1 with TabICL v2 (exploratory)

[TODO: G8_REPORT §2, §6.4]

D Study 2: sensitivity analysis and full results

Table 3: Study 2: mean score of every method (datasets averaged over 3 seeds). PRODEN (default) failed on gas-drift in 5 of 6 cells (non-finite probabilities, reproduced on a rerun); its $q = 0.3$ mean is over 9 datasets and its $q = 0.6$ gas-drift entry uses one seed.

LLP, bag 8 (AUC, 18 datasets)			PLL (accuracy, 10 datasets)				
Method	default	tuned	Method	$q = 0.3$		$q = 0.6$	
				default	tuned	default	tuned
In-Context Gibbs (ours)	0.840		**In-Context Gibbs (ours)**	0.862		0.851	
Hard-EM (same TFM)	0.827		Hard-EM (same TFM)	0.862		0.803	
Stochastic EM (same TFM)	0.835		Stochastic EM (same TFM)	0.858		0.843	
Naive TFM	0.724		Naive TFM	0.840		0.647	
DLLP (MLP)	0.783	0.799	PRODEN	0.800	0.836	0.678	0.820
DLLP (linear)	0.788	0.794	GLWS	0.814	0.848	0.724	0.837
LLP-VAT	0.778	0.798	RC	0.828	0.849	0.803	0.839
EasyLLP	0.724	0.771	CC	0.827	0.850	0.804	0.839
GLWS	0.784	0.789	Oracle (true labels)	0.870		0.870	
Oracle (true labels)	0.879						

[TODO: S1 sensitivity (S2_RESULTS §1, §2), per-baseline comparisons (§4), per-dataset tables]

E Datasets, baselines and tuning spaces

[TODO: dataset cards; docs/tuning_spaces.md]

F Secondary settings of Study 1

[TODO: LLP bags 16/32, PU-SCAR, PU-SAR, noisy labels, SSL; D1-D3]